

HAT-P-37b

R-2606

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_250725 · created 2026-08-28T01:07:13+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460878.942 +/- 0.013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.048
Transit depth (Rp/Rs)^2	2.89 +/- 1.63 [%]
Orbital Inclination (inc)	89.44 +/- 2.52
Airmass coefficient 1 (a1)	1.179 +/- 0.02
Airmass coefficient 2 (a2)	-0.15 +/- 0.21
Scatter in the residuals of the lightcurve fit is	1.69 %
Best Comparison Star	#1 - [428, 434]
Optimal Aperture	3.43
Optimal Annulus	15.18
Transit Duration (day)	0.056 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.17 $\pm$ 0.048 vs 0.1394 (deviation 0.64 σ)
Duration fit vs published	0.056 d vs 0.0983 d (deviation 3.85 σ)
Mid-time O-C	-4047.3 min
Depth SNR	3.5
Residual scatter	1.69 %

Light curve

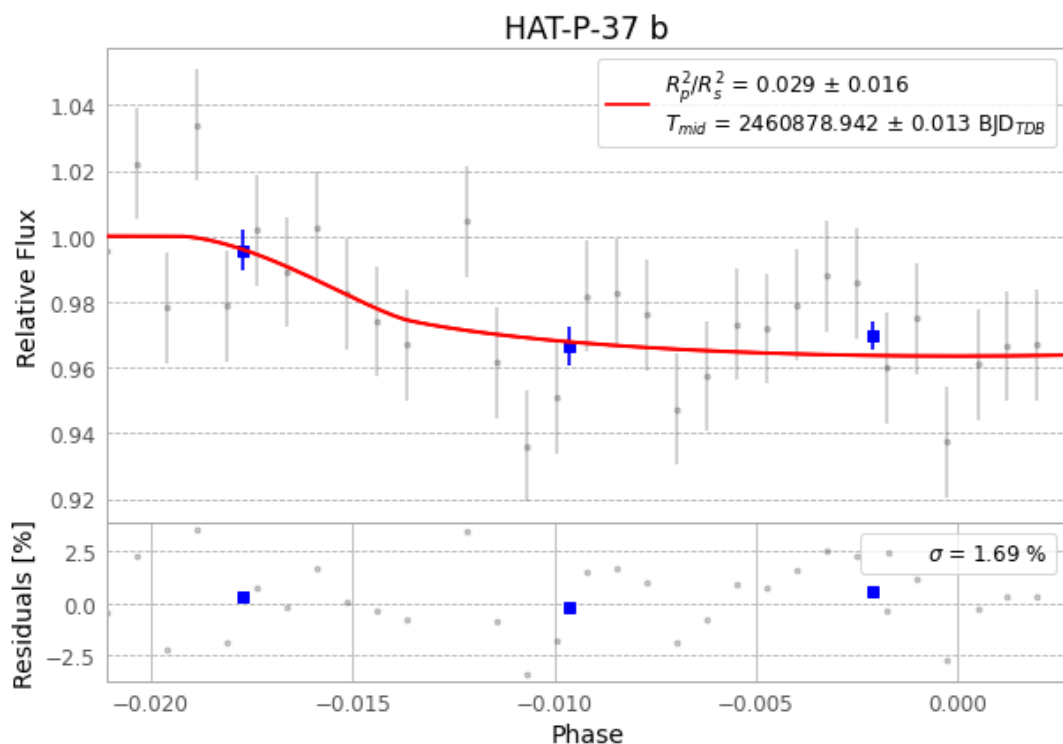


Figure 1 — FinalLightCurve_HAT-P-37 b_2025-07-24.png (SHA-256 926ebcb6a46e4941...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [336, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2e6b88f3ba711deb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

33 FITS frames (22 MB), HATP-37250725041515.FITS → HATP-37250725055115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-250725.log (964bd8546ce1...), FinalLightCurve_HAT-P-37 b_2025-07-24.png (926ebcb6a46e...), FinalLightCurve_HAT-P-37 b_2025-07-24.csv (8209a5c67077...)

Compute resources

Wall time 153.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 01:07Z.